

Visualizing the Hawai‘i Electric Grid

A data-focused technical report on the SCEL solar data visualization project
(O‘ahu renewable share, modeled duck curve, curtailment, battery simulation and plant
map)

September 2026

Data refreshed 20 September 2026; screenshots and figures show the site as of that date

Abstract

This report documents a small, open, reproducible data-visualization project about the electric grid of O‘ahu, Hawai‘i: a Python pipeline that pulls public data (U.S. Energy Information Administration, Hawaiian Electric, and the National Renewable Energy Laboratory), caches it, processes it, and writes static JSON files that five interactive web pages display. The report concentrates on the *data*: what each source measures, how fresh and complete it is, how the numbers are combined, where the results are measured and where they are modeled, and what the charts do and do not show. It also explains the relevant power-systems and solar concepts, proposes additional data that would make the visualizations more useful (and to whom), and gives a candid assessment of where the site is novel, where it is not, and who it serves. Headline results from the current data: utility-scale solar and wind supplied 11.1% of O‘ahu’s generation in 2025 (0% until 2011), roughly 26% of all electricity used when an estimate of rooftop solar is added; wind and solar curtailment was 1.6% of what the plants could have delivered in 2025 (7.1% in the worst year, 2021), 95% of it recorded as “system constraint”; and a modeled spring weekday has a 387 MW midday net-load minimum and a 993 MW evening peak.

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1 Executive summary

What the project is

A static website (no server-side code) with five pages, fed by a four-stage Python pipeline (fetch → cache → process → export). It is built for O‘ahu first, with a **region** field on every record and per-island settings in one configuration file so that other islands can be added. Every chart is labeled by how its numbers were obtained: **Reported** (a data provider published it), **Modeled** (we computed it from shapes and totals), or **Simulation** (a what-if).

Page	Nature of the numbers	Question it answers
Renewable share (Tier 1)	Reported, plus an estimate of rooftop solar	How much of O‘ahu’s electricity has come from utility-scale solar and wind since 2001, and what changes if rooftop solar is counted?
Duck curve (Tier 2)	Modeled	What does a typical weekday look like hour by hour once solar is on the grid: how deep is the midday dip and how steep is the evening climb?
Curtailment (Tier 3)	Reported (Hawaiian Electric)	How much wind and solar could have been used but was held back, and why?
Battery (Tier 4)	Simulation on the modeled day	What would batteries do to the evening peak, the midday low and curtailment, for a chosen battery size and amount of extra solar?
Plant map	Reported (EIA-860M inventory)	Which plants and units existed, where, and when did they start or retire (1947 to 2029, including planned units)?

Table 1: The five pages of the site.

What the data say (current refresh)

- **Renewable share.** Utility-scale solar plus wind supplied 0% of O‘ahu’s generation from 2001 to 2010, 0.8% in 2011 (the year Kahuku Wind started), 8.5% in 2020, 12.0% in 2023 and 11.1% in 2025 (2025 is preliminary). The best single month was June 2023, at 15.3%. Adding an estimate of rooftop solar and counting it in both the numerator and the denominator, the share of *all* electricity used rose from 8.1% (2014) to 26.6% (2025).
- **The fuel mix is still overwhelmingly oil.** In 2024 (final data), residual and distillate fuel oil supplied 82.6% of the 6,374 GWh generated by the utility-scale plants, solar 7.1%, wind 4.7%, waste-to-energy 5.3%.
- **Curtailment.** Hawaiian Electric reports that it held back 14.3 GWh of wind and utility-scale solar in 2025 (1.6% of the 875.6 GWh potential), against a peak of 40.9 GWh (7.1%) in 2021. In 2025, 95% of it was classified as “system constraint” rather than “oversupply”; the oversupply category fell from 22.0 GWh in 2021 to 0.6 GWh in 2025.
- **The modeled duck.** On a spring weekday the modeled net load (what fuel-burning plants must supply) bottoms out at 387 MW at noon, peaks at 993 MW at 6 PM, and climbs 606 MW in between; the steepest three-hour ramp is +397 MW starting at 2 PM.
- **Storage.** O‘ahu has six batteries totalling 326.4 MW and 1,124.9 MWh (after correcting one implausible EIA figure). Treated as a single battery on the modeled spring day, the fleet would lower the evening peak from 993 to 687 MW—an upper bound that assumes it may charge from the grid whenever it likes.

- **Inventory.** 32 plants (2,394 MW of nameplate capacity, of which 1,429 MW oil-fired, 355 MW solar, 127 MW wind, 158 MW waste-to-energy and 326 MW batteries) are in service as of the July 2026 EIA file; five plants (18 units, 191 MW) are listed as planned through April 2029.

Assessment in one paragraph

None of the underlying data is new: everything comes from public sources. The contribution is integration and clarity: one place where the island’s generation history, rooftop estimate, curtailment record, battery fleet and plant timeline are put on consistent definitions, with the provenance of every number stated and a fully documented, tested, re-runnable pipeline. That makes the site genuinely useful for teaching, for non-specialist audiences (students, journalists, community and policy staff) and as a research starting point. It is *not* an operational tool: the duck curve is modeled rather than measured, curtailment is only available quarterly and combines wind with solar, and the battery simulation is a stylized optimizer. The most valuable next step is not more charts but better data (Section 8): hourly system and curtailment data from the utility would turn the modeled pages into measured ones.

2 Background: solar and power-systems concepts used in this project

This section gives the minimum vocabulary needed to read the charts; a glossary follows in Appendix A.

2.1 Power versus energy, capacity versus generation

Power is a rate, measured in watts (W), kilowatts (kW), megawatts (MW). Energy is power sustained over time, in watt-hours: one megawatt for one hour is one megawatt-hour (MWh); a thousand MWh is a gigawatt-hour (GWh). A power plant’s *nameplate capacity* (MW) is the most it can deliver; its *generation* (MWh) is what it actually delivered over a period. The ratio of generation to the maximum possible (capacity $\times$ hours) is the *capacity factor*. Solar has a capacity factor well below one because the sun sets and clouds pass; oil-fired plants can in principle run all hours but on O’ahu are dispatched to follow demand. The map page shows *capacity*; the renewable-share and curtailment pages show *energy*. Confusing the two is the most common misreading of energy charts, so each page says which one it shows.

2.2 Utility-scale versus distributed (rooftop) solar

Utility-scale solar is a farm connected to the grid and dispatched, in some sense, by the utility. *Distributed* or *behind-the-meter* (BTM) solar is on customers’ roofs: it first reduces what the customer draws from the grid, and any surplus flows out. The utility does not see the panels’ output directly; it sees the *net* demand of the customer. This matters because (a) rooftop generation is not in plant-level generation data such as EIA-923, and (b) rooftop solar cannot normally be curtailed by the utility, so when there is too much solar, the utility can only turn down the utility-scale plants. On O’ahu rooftop solar is large: Hawaiian Electric reports 1,211 GWh of customer-owned renewable generation on O’ahu in 2024, which is more than two and a half times the 455 GWh from utility-scale solar that year.

2.3 Balancing an island grid

O‘ahu’s grid is not connected to any other grid. Supply and demand must therefore be matched second by second within the island, using the island’s own generators. Operators keep some units running at minimum output (“must-run”) for voltage support, spinning reserve and *inertia*: the rotating mass of large synchronous generators that resists sudden frequency changes. Solar panels and batteries connect through inverters and do not, by default, provide inertia; newer “grid-forming” inverters can. Fossil units also have a *minimum stable output* below which they cannot be turned down without shutting off, and *ramp-rate limits*. These constraints, not just the total amount of sun, decide how much solar can be accepted.

2.4 Net load and the duck curve

Net load is demand minus the output of variable renewables, i.e. what dispatchable (here mostly fuel-burning) plants must supply. On a sunny day, midday solar pushes net load down; when the sun sets just as people return home, net load climbs steeply to the evening peak. Plotted over 24 hours the shape resembles a duck (the “belly” is the midday dip, the “neck” the evening climb); the shape was popularized by the California ISO around 2013 [9]. The relevant engineering questions are three numbers per day: how low the midday minimum goes (does it approach the minimum stable output of the fossil fleet?), how high the evening peak is (how much firm capacity is needed?) and how fast the climb is (how quickly must units ramp?). Tier 2 reports exactly these three.

2.5 Curtailment

Curtailment is an instruction to a generator to produce less than it could. For wind and solar this wastes free energy, and it affects economics because energy producers under power-purchase agreements (PPAs) are typically paid per MWh delivered. Hawaiian Electric classifies curtailed energy by reason (see Section 4.4): *oversupply* (more supply than the whole system can use), *system constraint* (limits of particular equipment or parts of the grid) and *facility requested* (the plant’s owner asked). The three reasons call for different remedies: storage and demand shifting address oversupply, whereas transmission or inverter-control upgrades and operating-practice changes address constraints. That is why the breakdown is more informative than the total.

2.6 Storage

A battery has a power rating (MW: how fast it can charge or discharge), an energy capacity (MWh) and a round-trip efficiency (the fraction of energy put in that comes back out; typically in the mid-80% range for lithium-ion). The ratio of energy to power is its duration in hours; O‘ahu’s fleet is about 3.5 hours. Batteries can perform *load shifting* (charge at midday, discharge in the evening, which is what Tier 4 simulates) and also *ancillary services*: fast frequency response, synthetic inertia and black start, none of which the simulation models.

2.7 Why Hawai‘i is a useful place to study this

Hawaiian Electric serves O‘ahu (together with Maui County and Hawai‘i Island) and Hawai‘i law, since Act 97 of 2015, sets a target of 100% renewable electricity by 2045, the first such state law [11]. O‘ahu combines several features rarely found together: an isolated grid, a generation fleet historically dependent on imported oil (with a coal plant, AES Hawai‘i, that closed on 1 September 2022 [13]), very high rooftop solar adoption, helped by high electricity prices

and net-metering policy (closed to new customers in October 2015 [12]), and a large grid-scale battery (Kapolei Energy Storage, December 2023 [14]). It is a natural preview of the integration problems other systems will face at higher renewable shares.

3 System architecture

3.1 Design goals

The project was built to a small set of standing rules: (1) each pipeline stage in its own module, so a source or calculation can change without touching the rest; (2) the raw downloads are cached separately from derived results, so processing can be re-run with no network; (3) every run is idempotent (running it twice changes nothing); (4) every record carries a **region** field, and each region’s settings (its list of plants, its county, its calibration constants) live in `config/regions.yaml`; (5) API keys live only in a git-ignored `.env`; (6) the web pages read pre-generated JSON, never live APIs; (7) any change to how an existing number is calculated must be approved, as opposed to how it is displayed or explained.

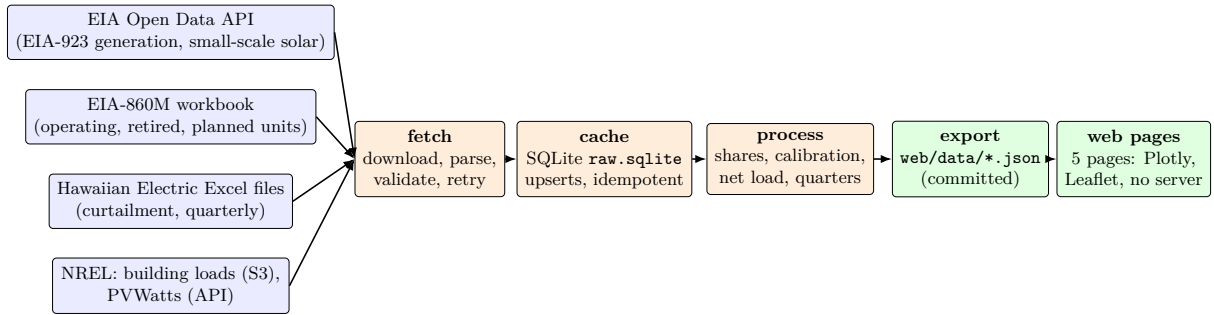


Figure 1: Data flow. Each arrow is a separate module; the cache lets the process stage re-run offline. The battery simulation has no pipeline of its own: it runs client-side on the modeled JSON.

3.2 Modules

- **Fetch** (`pipeline/fetch*.py`). One module per source. They download, retry on transient errors, and *validate what came back*. Examples of why this matters: EIA answers a request for a not-yet-published monthly workbook with a normal-looking HTML page and a “200 OK” status, so the fetcher checks that the bytes are a real Excel (zip) file; Hawaiian Electric’s Excel files are made for people, so rows are found by their labels rather than fixed cell positions, and the fetcher stops with a clear error if the layout changes rather than reading the wrong cells.
- **Cache** (`pipeline/cache.py`). SQLite tables named `raw_*` hold exactly what was downloaded. Rows are upserted on a natural key, so re-running never duplicates. Derived numbers are never stored here.
- **Process** (`pipeline/process*.py`). Pure functions: rows in, dictionaries out. No network, files or database, so they are unit-tested directly.
- **Export** (`pipeline/export*.py`). Writes the JSON the pages read and stamps it with a generation time, source description, and kind (reported or modeled).
- **Web** (`web/`). Static HTML, CSS and JavaScript modules. Plotly draws the charts and Leaflet (with OpenStreetMap tiles [15]) draws the map. There is no build step and no

server-side code: the pages can be hosted on any static host, and the only network calls at view time are the JSON files, the chart libraries and the map tiles.

3.3 Testing

The repository has three layers of automated tests. 109 Python tests cover fetch parsing (including the quirks of the real files), cache idempotence, every calculation, corrections, and the hand-written milestone file (whose EIA-derived dates are checked against the generated plant data). 13 JavaScript tests (run with Node) test the battery optimizer directly: power and energy limits are never exceeded, energy is conserved through the efficiency losses, a flat day is left alone, a larger battery is never worse, and it runs in tens of milliseconds. A headless-browser suite drives the real pages through the Chrome DevTools Protocol: it clicks the buttons, drags the sliders, hovers and clicks on chart points with a real mouse, and checks the layout at phone width. The browser suite caught real bugs that screenshots had not, for example the “5y” and “10y” zoom buttons showing the wrong years because Plotly had padded the time axis.

4 The data sources in detail

4.1 EIA-923: what each plant generated

The core of Tier 1 is the generation reported by power plants to EIA on Form EIA-923 [2], read through EIA’s Open Data API [1]. A *region* is defined as a list of plants, because EIA does not publish a fuel mix per utility or island, only per plant. The 38 plants listed in the configuration form the O’ahu definition; they were found by matching EIA’s plant registry by county (Honolulu County is the island of O’ahu) and include independent power producers selling to Hawaiian Electric, plus retired plants (Honolulu, AES Hawai’i) so that the history is complete. Two more plants in the same county, refinery cogeneration plants (Tesoro/Par Hawai’i and Hawaii Cogen), are excluded, with the reason recorded in the configuration, because they power the refinery rather than the grid.

EIA publishes each plant’s numbers several overlapping ways (a plant total, a per-fuel breakdown, a per-fuel-and-engine-type breakdown). The pipeline requests only the per-fuel level and drops plant-total rows so nothing is counted twice. Battery charging appears in the data as negative generation from the fuel code for storage, and is excluded from the totals because storage is not generation.

Provisional months. About a dozen small O’ahu plants (mostly solar farms) report to EIA only once a year. Until they do, the most recent months are missing their output and read low—by roughly 1 to 1.5 percentage points in recent years. The pipeline detects the last month for which the annual reporters’ data exists (their data ends in a December cluster) and marks later months **provisional**; the chart shades them and the table labels them “preliminary”. Once EIA publishes the missing year the next run corrects the numbers by itself. For the current refresh, data are final through December 2024, so all of 2025 and 2026 are preliminary, and the 2025 renewable-share figures in this report carry that caveat.

4.2 The rooftop-solar estimate and how it is anchored to Hawaiian Electric

EIA-923 does not cover rooftop panels, so the solid line in Tier 1 understates renewables. EIA publishes its own estimate of small-scale solar generation, but for the whole state and only from 2014. The dotted line brings it down to O’ahu. An earlier version multiplied the state figure

Source	What it measures	Resolution	Freshness	Main limits
EIA Form EIA-923 (via the EIA Open Data API, route <code>electricity/ facility-fuel</code>)	Net generation and fuel by plant and fuel type, in MWh	Monthly (about 3,000 U.S. plants); annual only for about 9,500 smaller plants	About 2–3 months behind; final figures about a year later	Plants of 1 MW or more only; recent months are “provisional” because annual-only plants report late; no hourly data
EIA small-scale solar estimate (API route <code>electric-power- operational-data</code>)	EIA’s <i>estimate</i> of generation from solar systems under 1 MW, mostly rooftop	Monthly, statewide, from 2014	A few months behind	Whole state of Hawai‘i, not O‘ahu; an estimate from utility net-metering reports, not a meter read- ing
EIA-860 / EIA-860M (Excel; Operating, Retired and Planned sheets)	The inventory of generating units: size, fuel, technology, location, in-service and retirement month; storage energy (MWh); planned units with expected month	One row per generator; monthly file	Monthly, about a month behind	Plants ≥ 1 MW total and grid-connected; retire- ments listed only from 2002; storage energy occasionally implausible; planned dates are the developers’ expectations
Hawaiian Electric curtailment workbooks (Key Performance Metrics)	Delivered and curtailed wind plus utility-scale solar energy by island, and (since mid-2015) curtailed energy by reason	Quarterly, from Q1 2012	Quarterly; past values are sometimes corrected	Wind and solar combined; no hour or site detail; the definition of each reason is not published with the data
NREL End-Use Load Profiles (ResStock, ComStock; public S3 bucket)	Simulated electricity use of homes and businesses, and simulated output of rooftop panels on homes	15-minute, using actual 2018 weather	A fixed simulated year	A model, not a meter reading; time-stamped in Eastern Standard Time (5 hours off from Hawai‘i); homes are statewide files, businesses are by county
NREL PVWatts v8 (API)	Hourly output of a reference solar farm at a given location, in a typical meteorological year	Hourly	Static	A typical year, not 2024 weather; only the daily <i>shape</i> is used

Table 2: Data sources, what they measure and their limits. All are public and free; only the EIA and NREL APIs need a (free) key.

by a fixed 77%, the ratio of Hawaiian Electric’s reported 2024 customer-owned generation on O’ahu (1,211 GWh) to EIA’s estimate for the state (1,573 GWh). But the measured ratio varies from about 0.65 (2017) to 0.83 (2022), so a constant would be off by up to roughly 18% in some years. The current method anchors each quarter to Hawaiian Electric’s own reported O’ahu total:

$$R_m = C_q \frac{S_m}{\sum_{m' \in q} S_{m'}} \quad (m \in q), \quad (1)$$

where S_m is EIA’s statewide small-scale solar estimate for month m , C_q is Hawaiian Electric’s reported customer-owned generation for the quarter q containing m , and R_m is the resulting O’ahu estimate. Each quarter therefore adds up exactly to the utility’s number (this was checked for all 50 quarters), and the state series decides only how a quarter is split across its three months. Where a quarter has not been reported, or the state quarter is incomplete, the fixed 77% share is used and the record’s `rooftop_basis` field says “assumed”; in the current data all 150 months with an estimate are “measured”. Because rooftop energy is added to both the numerator and the denominator, the dotted line is

$$s_m^+ = 100 \frac{E_m^{\text{ren}} + R_m}{E_m^{\text{tot}} + R_m}, \quad (2)$$

the share of *all* electricity used, not just what utility-scale plants generated. The two lines therefore measure different things and are labeled as such. Adding the dotted line changed no existing number (a before/after comparison of the JSON confirmed this).

4.3 EIA-860M: the plant inventory

Form EIA-860 [3] is EIA’s inventory of the country’s power plants. A plant must report if two conditions hold: its generators sum to at least 1 MW of nameplate capacity, and it is connected to the local or regional electric grid. The plant’s operator files it, annually with a monthly update (EIA-860M). For each generator it records capacity, fuel, technology, location, in-service month, retirement month (once retired), and for batteries the energy capacity. This gives three uses in the project: the plant map, the battery fleet used as the Tier 4 default, and the planned-unit view.

What it *cannot* tell us matters as much. Rooftop and other small systems are absent; generators not connected to the grid are absent; and EIA’s monthly file lists retirements only from 2002, so a plant that closed earlier does not appear at all. That is why the map shows a single unit in 1947 (a Waiau unit retired in 2024, the only unit from that era EIA still lists), and why the 1980s Kahuku wind turbines, the last of which shut down in 1996, are not shown. The page explains this in a collapsed explainer rather than adding data by hand, because the requirement was “EIA-860 plants only”.

A documented correction. EIA lists the 36 MW battery at Waiawa Solar Power Hybrid as storing 4.6 MWh (a maximum discharge of about 2.3 MW), which is not plausible; its developer reports 144 MWh [10]. The configuration file records this correction with its source and a guard: it applies only while EIA still publishes exactly 4.6, so if EIA revises the value the pipeline warns rather than overriding a possibly correct number. The exported JSON keeps both the value as filed and the value used. The correction changes the fleet’s stored energy from 985.5 to 1,124.9 MWh.

Planned units. The Planned sheet lists units whose operators have told EIA they intend to build them, with an expected in-service month and a stage from “under construction, more than 50 percent complete” to “planned, regulatory approvals not initiated”. For O’ahu, it lists

18 units at five plants; the furthest is Puuloa Energy (eleven 9.4 MW waste-biomass units, 103 MW, expected April 2029, at the earliest stage of approval). The map treats them as planned, never as in service. Two facts limit this view: expected dates are the developers’ and often slip, and EIA lists *no planned retirements* for O’ahu, so the future view can only add plants, never remove one.

4.4 Hawaiian Electric’s curtailment files

Hawaiian Electric publishes two Excel files: delivered versus curtailed energy per island and quarter (from Q1 2012), and curtailed energy by reason (from July 2015). The original plan for Tier 3 was to *estimate* curtailment by comparing a modeled “potential” solar output with actual generation. That was abandoned because the utility publishes the measured figures, and because since 2015 curtailment has been only 1–7% of potential a year, smaller than the uncertainties in the inputs a model would need (DC-to-AC capacity ratio, actual versus typical weather, panel aging). A modeled estimate could not have resolved what is measured directly. (This is reasoning, not a tested result; the modeled version was not built.)

Reading the files. “Delivered” plus “curtailed” is “potential”: what the plants could have supplied. The categories are wind and utility-scale solar *together* (“curtailable renewable resources”), so the page cannot say how much was solar. Rooftop solar is not curtailable and is not part of potential. Annual totals are sums of the quarters. Two internal inconsistencies are worth reporting: Hawaiian Electric’s own annual figures differ slightly from the quarter sums for a few years after corrections described in its notes, and the two files disagree with each other in places: the by-reason columns sum to about 1.0 GWh *more* than the totals file in 2018 (6.40 versus 5.40 GWh) and about 1.5 GWh more in 2023 (23.98 versus 22.49 GWh). The pipeline reports totals from the totals file and reason amounts from the by-reason file, and marks a year’s reason breakdown as complete only when the reasons cover the whole year; the “95%” headline for 2025 is the system-constraint amount divided by the totals file’s curtailed energy.

4.5 NREL building-load and solar-shape data

No public hourly grid data for Hawai’i exists that we could find: EIA’s Hourly Electric Grid Monitor covers the Lower 48 [4] and Hawaiian Electric publishes monthly and annual summaries. Tier 2 is therefore modeled: hourly *shapes* come from NREL simulations [7, 8], and hourly *sizes* are calibrated to the real monthly EIA data. Two technical gotchas are worth recording because they would have silently corrupted every curve. NREL’s load files are stamped in Eastern Standard Time (UTC−5) rather than Hawai’i time (UTC−10), and each stamp marks the *end* of a 15-minute interval; the fetcher shifts them 5 hours earlier, and a test guards this—without it the “belly” would land at the wrong time of day. Second, residential load files are available only statewide, so O’ahu’s share of homes (0.70, approximated from its share of the population in the 2020 Census) sets the residential part of demand.

5 Methods by tier

5.1 Tier 1: renewable share

For each month, with $g_{p,f,m}$ the generation of plant p from fuel f :

$$E_m^{\text{tot}} = \sum_p \sum_{f \notin \text{storage}} g_{p,f,m}, \quad E_m^{\text{ren}} = \sum_p \sum_{f \in \{\text{solar, wind}\}} g_{p,f,m}, \quad s_m = 100 \frac{E_m^{\text{ren}}}{E_m^{\text{tot}}}. \quad (3)$$

Only solar and wind count as renewable (a project definition); waste-to-energy (H-POWER) and biomass do not. This is narrower than EIA’s own “renewable” category and than the statutory definition used for Hawai‘i’s renewable portfolio standard, so the percentages here are *not* compliance figures. The JSON also keeps generation by every fuel each month so that other views can reuse it.

5.2 Tier 2: the modeled typical day

Let h be the hour and m a month of the reference year (the latest year of final Tier 1 data; 2024). Each component has an hourly *shape* $f_{m,h}^{(c)}$ from NREL and a monthly *size* $M_m^{(c)}$ (MWh) from EIA, and the hourly series is $x_{m,h}^{(c)} = M_m^{(c)} f_{m,h}^{(c)} / \sum_h f_{m,h}^{(c)}$. The components and their sizes are: utility solar (PVWatts shape; size from EIA-923 solar), wind (flat shape; size from EIA-923 wind), rooftop solar (ResStock shape; size R_m as in Section 4, using the fixed 77% share for the reference year, where it is exact) and customer demand (ResStock plus ComStock shape; size equal to O‘ahu’s generation plus rooftop solar, i.e. an energy balance ignoring losses and imports). In megawatts, hour by hour:

$$\text{grid load}_h = \text{customer demand}_h - \text{rooftop}_h \quad (\text{what the utility “sees”}) \quad (4)$$

$$\text{net load}_h = \text{grid load}_h - \text{utility solar}_h - \text{wind}_h \quad (\text{what fuel-burning plants supply}) \quad (5)$$

A “typical weekday” in each season is the hourly average of that season’s Monday–Friday days (64–66 weekdays per season). Four headline numbers are computed from the net-load curve: the midday low (minimum over 9 AM to 3 PM), the evening peak (maximum after it), the climb between them, and the steepest three-hour ramp.

Season (weekday average)	Midday low	Evening peak	Climb	3 h ramp
Winter (Dec–Feb)	490	1,072	+582	+414
Spring (Mar–May)	387	993	+606	+397
Summer (Jun–Aug)	460	1,017	+557	+376
Fall (Sep–Nov)	538	1,064	+526	+359

Table 3: Modeled net-load statistics for a typical weekday, reference year 2024, all in MW (“3 h ramp” is the steepest three-hour ramp). The midday low is at noon and the peak at 6 PM in every season; the steepest ramp starts at 2 PM.

Validation, honestly. What could be checked, and what could not:

- *Rooftop total.* The model’s rooftop energy for 2024 (1,211 GWh) equals Hawaiian Electric’s reported figure. This is agreement *by construction*, since the 77% share was derived from that same figure; it checks the arithmetic, not the model.
- *Evening peak (an independent check).* Hawaiian Electric told the Hawai‘i Public Utilities Commission that O‘ahu’s 2020 system peak was 1,087 MW at 5:26 PM, with 2010–2019 peaks between 1,141 and 1,206 MW [6]. The modeled average weekday peaks at 993–1,072 MW around 6 PM, depending on season: the right size and nearly the right hour, and, as expected, a little below the single busiest day.
- *Delivered energy.* Hawaiian Electric’s “delivered” wind plus solar is 0.85–0.94 of EIA’s solar plus wind for 2021–2024. The two count slightly different plant sets, so exact equality is not expected; a stable ratio near one shows that both describe the same quantity.

- *Not checked.* The overnight and midday minima, for which no public figures were found. The modeled overnight grid load (roughly 40–50% of the evening peak) looks lower than would be expected for O‘ahu, which exaggerates how deep the belly looks relative to the night.
- *Not modeled.* Curtailment, so the true midday dip is probably shallower than drawn.

Structurally, the chart combines 2018 weather shapes with 2024 sizes, uses one average weekday per season (which smooths out cloudy and clear days), and leaves out weekends. It illustrates the pattern; it does not record any day.

5.3 Tier 3: curtailment

No modeling is involved. The pipeline reads the two workbooks, locates O‘ahu’s section by label, and computes $\text{potential} = \text{delivered} + \text{curtailed}$ and $\text{curtailment \%} = 100 \text{ curtailed} / \text{potential}$ for each quarter and year, with reasons where available. Cross-checks against Tier 1 (the ratio above) and against EIA’s rooftop estimate (which fixed the 77% share) are printed on every run.

5.4 Tier 4: the battery simulation

For a 24-hour net-load curve n_h , battery power limit P , energy capacity B and round-trip efficiency η (split evenly, $\sqrt{\eta}$ each way), the browser chooses a charge/discharge schedule to minimize, over a run of $T = 120$ hours (the same day repeated five times),

$$\sum_{t=1}^T \left[\max(r_t, F)^2 + \lambda \text{curt}_t + \epsilon (c_t + d_t) \right], \quad r_t = n_t + c_t - d_t, \quad \text{curt}_t = \max(0, F - r_t), \quad (6)$$

where c_t and d_t are the power drawn from and delivered to the grid, subject to $c_t, d_t \leq P$ and a state of charge kept in $[0, B]$; F is the “fossil minimum output” floor; $\lambda = 4 \max(F, \max_h n_h) + 1$ is a penalty large enough that absorbing one MW of surplus always beats any flattening it could buy; and $\epsilon = 0.01$ is a tiny throughput penalty that only breaks ties. The squared term models fossil plants becoming costlier per MWh as they are pushed harder, so minimizing it flattens the day. It is solved exactly by dynamic programming over a 200-level state-of-charge grid, with perfect foresight. The battery starts *empty* and the third of the five days is shown, so the display is a steady-state day. (Starting with a free charge was tried first and was wrong: the optimizer simply spent the gift slowly, so the “typical day” was a draining day. A test now guards this.)

The sliders are battery power (default 326 MW, the island’s fleet), energy (1,125 MWh), efficiency (85%), extra utility-scale solar (0%) and the fossil minimum ($F = 300$ MW). The last is an **invented illustration**: no public figure for how low O‘ahu’s fossil units can be turned down was found; 300 MW was chosen so today’s typical weekday is not curtailed while extra solar visibly is, and it is labeled “illustrative” on the page.

What this simulation gets right and wrong. It models load shifting only. It does not model frequency response, synthetic inertia or black start; battery aging; reserves; market rules; weekends; demand response; line limits; or changes to rooftop solar. It also models only one kind of curtailment (oversupply against a floor), whereas Tier 3 shows that most recent curtailment is classified as system constraint, so it may *understate* what real batteries do. The optimizer also cycles the battery overnight (minimizing squared net load flattens the whole day), and energy stored outside solar hours comes from fossil plants, so it is not free.

Spring weekday, MW	Evening peak	Midday low	3 h ramp
No battery	993	387	+397
Kapolei alone (185 MW / 565 MWh)	808	—	—
Whole fleet (326 MW / 1,125 MWh)	687	557	+129

Table 4: Simulated effect of storage on the modeled spring weekday. The fleet result assumes the whole fleet may charge from the grid at will, which is an upper bound (batteries paired with solar farms may be restricted).

6 What the visualizations show

6.1 Renewable share over time

Table 5 and Figure 2 summarize Tier 1. The utility-scale share is zero for a decade, steps up with Kahuku Wind (2011), climbs slowly with small solar and Kawaihoa Wind, then jumps in 2020, the first full year of the four solar farms that started in late 2019 (about 130 MW; a smaller total in the pandemic year also lifts the ratio slightly). It plateaus at 11–12% from 2023. The estimate that includes rooftop solar tells a different story: it starts at 8.1% in 2014 and reaches 26.6% in 2025, because rooftop generation grew from an estimated 399 to 1,336 GWh a year. Roughly two thirds of the solar and wind electricity used on O‘ahu in 2025 (about 1,336 of 2,041 GWh) is thus rooftop solar, invisible to the plant-level survey, which is the main reason the site shows both lines.

Year	Total (GWh)	Solar+wind (GWh)	Share (%)	Rooftop (GWh)	Share incl. rooftop (%)
2005	8,163	0	0.0	—	—
2011	7,359	59	0.8	—	—
2014	7,109	208	2.9	399	8.1
2017	6,834	283	4.1	634	12.3
2019	6,779	298	4.4	881	15.4
2020	6,434	544	8.5	957	20.3
2022	6,440	655	10.2	1,101	23.3
2023	6,387	768	12.0	1,139	25.3
2024	6,374	753	11.8	1,211	25.9
2025*	6,331	705	11.1	1,336	26.6

Table 5: Selected annual values from the Tier 1 data. *Preliminary (final data end in December 2024): small annual-reporting plants are missing, so utility-scale figures read slightly low.

Fuel mix and capacity. In 2024 (final data) the utility-scale plants generated 6,374 GWh: residual fuel oil 5,091 GWh (79.9%), solar 455 (7.1%), wind 298 (4.7%), municipal solid waste 338 (5.3%), distillate fuel oil 174 (2.7%) and other biomass liquids 19 (0.3%). Nameplate capacity in service in July 2026 is dominated by oil (1,429 MW of 2,394 MW, 60%), while solar is 355 MW (14.8%). Because solar’s capacity factor is far below oil’s availability, a share of capacity translates into a much smaller share of energy.

A sanity check worth showing: implied capacity factor. Combining the two EIA sources gives a consistency check that the site does not currently display: generation from EIA-923 divided by nameplate capacity in service from EIA-860M times hours.

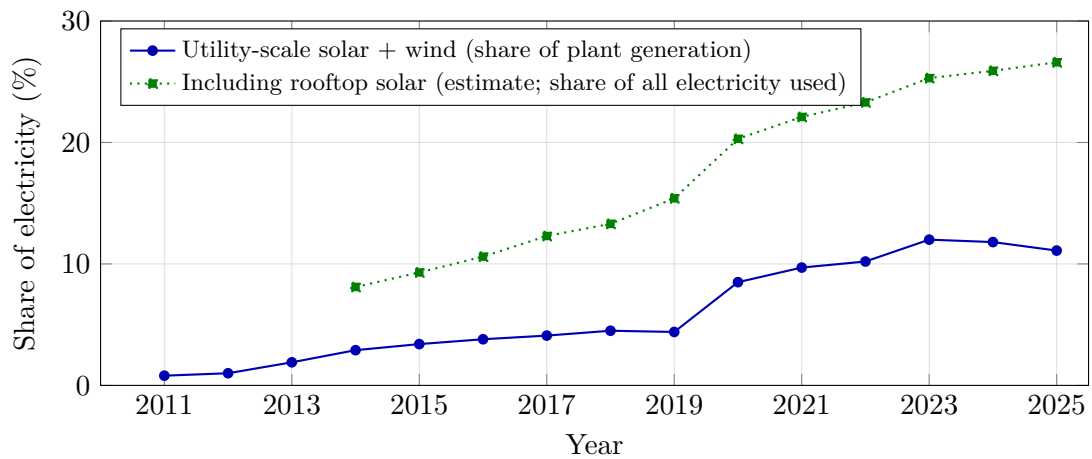


Figure 2: Annual renewable share, recomputed from the Tier 1 data (2025 preliminary). The two lines use different denominators (see Section 4).

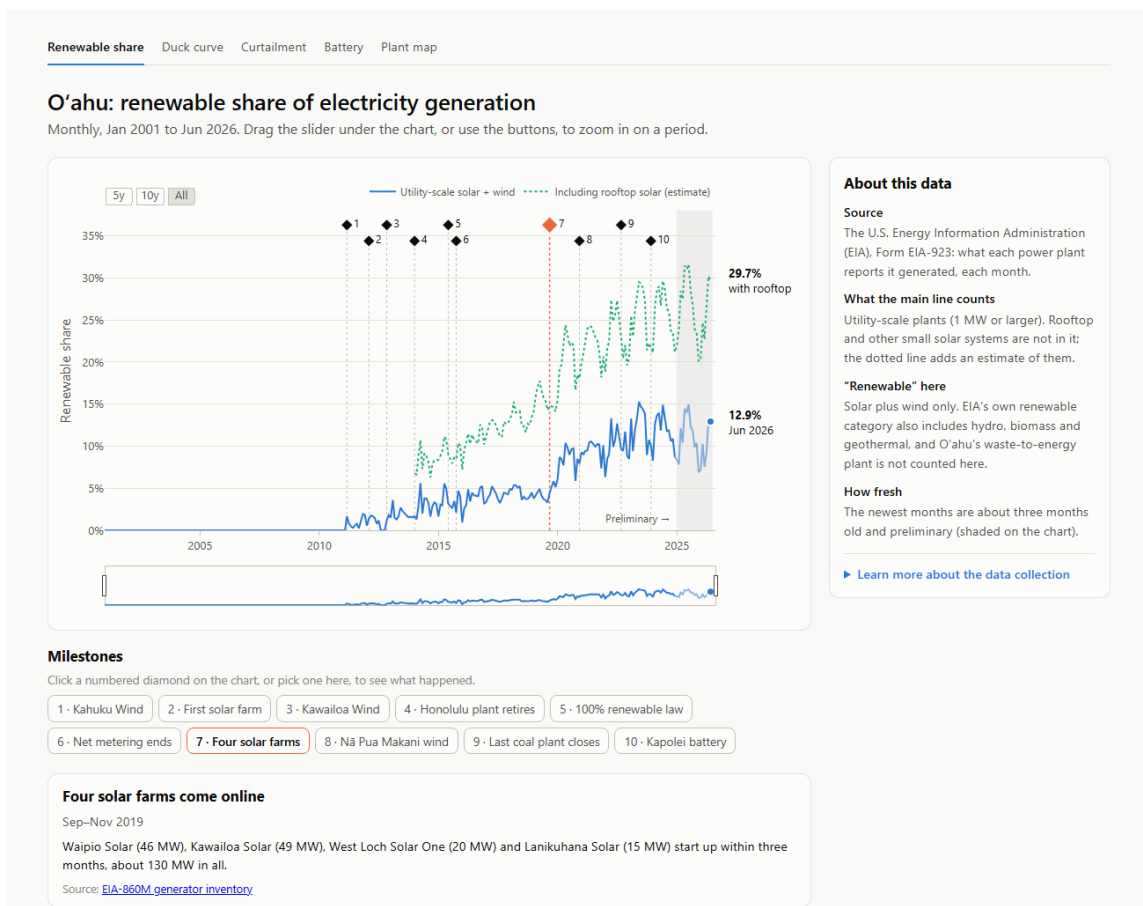


Figure 3: The Tier 1 page. Numbered diamonds are the ten milestones (e.g. 1: Kahuku Wind, March 2011; 7: four solar farms, September–November 2019, selected here); each opens a card with a one-sentence description and a source link. Hovering a line shows only that line's month and value.

	2019	2020	2022	2023	2024	2025*
Solar, average MW in service	95	195	222	277	287	322
Solar, implied capacity factor (%)	18.2	20.9	20.9	20.0	18.0	16.4
Wind, average MW in service	99	101	127	127	127	127
Wind, implied capacity factor (%)	17.0	20.9	22.5	25.6	26.8	21.9

Table 6: Implied capacity factors for the plants in the Tier 1 list, using each unit’s nameplate (AC) capacity from the month it enters service. *Preliminary generation. Values of 17–27% are plausible for O‘ahu, but the fall in solar’s figure in 2024–2025 is a question the data raises, not an answer: candidate explanations include new capacity ramping up mid-year, curtailment, weather, plant degradation, and (for 2025) missing small-plant data. The calculation is not part of the site.

6.2 The modeled duck curve

Figure 4 shows the modeled spring weekday: customer demand (dotted), grid load after rooftop solar, and net load after utility solar and wind. The rooftop layer is the largest source of the belly; utility solar and wind take a further bite. The evening climb of 606 MW over six hours from noon to 6 PM, with a steepest three-hour segment of +397 MW, is the number a system operator would translate into ramping requirements. Winter has the highest peak (1,072 MW) and fall the shallowest belly (538 MW).

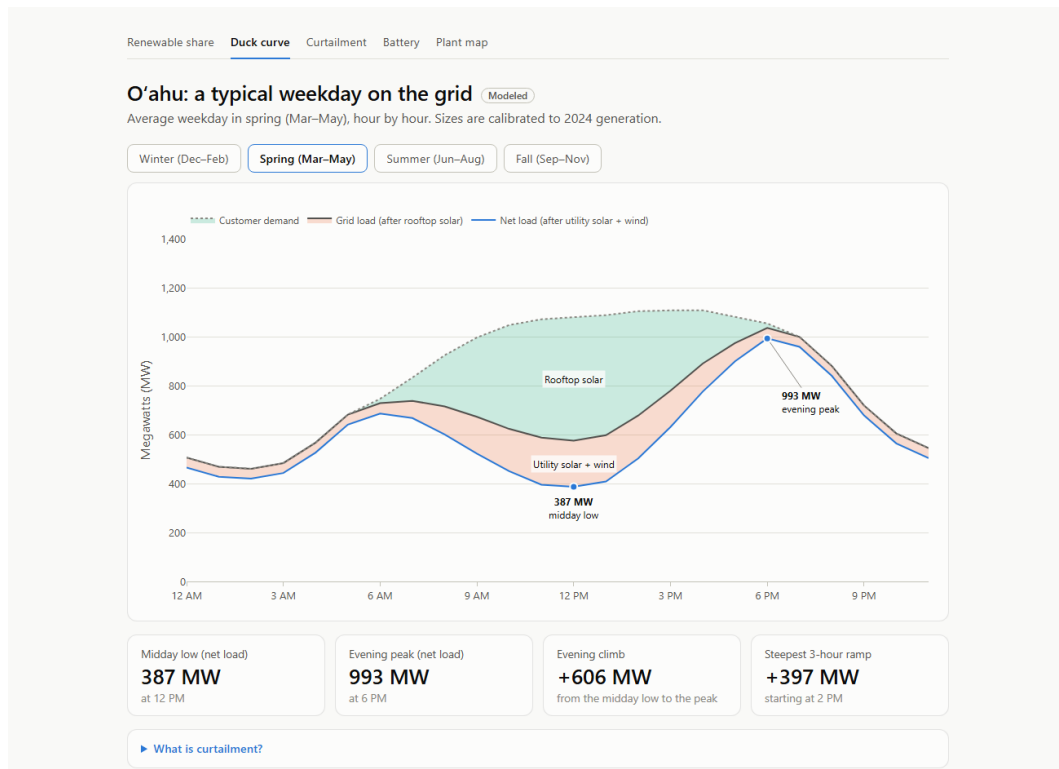


Figure 4: The Tier 2 page (spring, modeled). The page is tagged “Modeled” and says in words that it is an illustration of the pattern rather than a record of any day.

6.3 Curtailment

Curtailed wind and solar were small in energy terms but changed sharply in character. In 2012 it was 14% of a tiny base (Q3 2012 reached 25.7%); it fell to 1–2.4% from 2014 to 2019, rose to 7.1% in 2021 and 6.6% in 2022 as solar farms arrived, and dropped to 3.0% (2023), 3.5%

(2024) and 1.6% (2025). The reason breakdown (Figure 5) is more revealing than the total: from 2017 to 2019 nearly all curtailment was “system constraint”; in 2021–2022 about half was “oversupply”; and by 2025 oversupply had almost disappeared (0.6 of 14.3 GWh) while system constraint stayed around 14–19 GWh. The decline of oversupply curtailment coincides with new batteries, including Kapolei Energy Storage (December 2023), but the data alone cannot attribute it, since demand, solar additions, dispatch practice and weather also changed. And because system constraint is now the dominant category and Hawaiian Electric does not publish what it contains, the site cannot say how much of it storage could relieve. (An earlier draft of the project said batteries would help only partly; that was an overreach and was removed.)

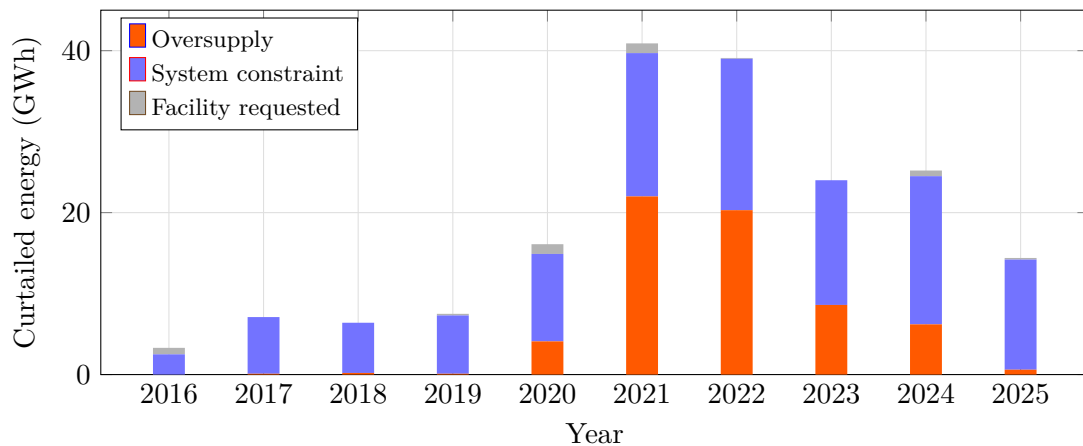


Figure 5: Curtailed wind and utility-scale solar on O‘ahu by reason, from Hawaiian Electric’s by-reason file (GWh per year; 2016 is the first full year of reasons). Bars show the by-reason sums; in 2018 and 2023 these exceeded the totals file by about 1.0 and 1.5 GWh (Section 4).

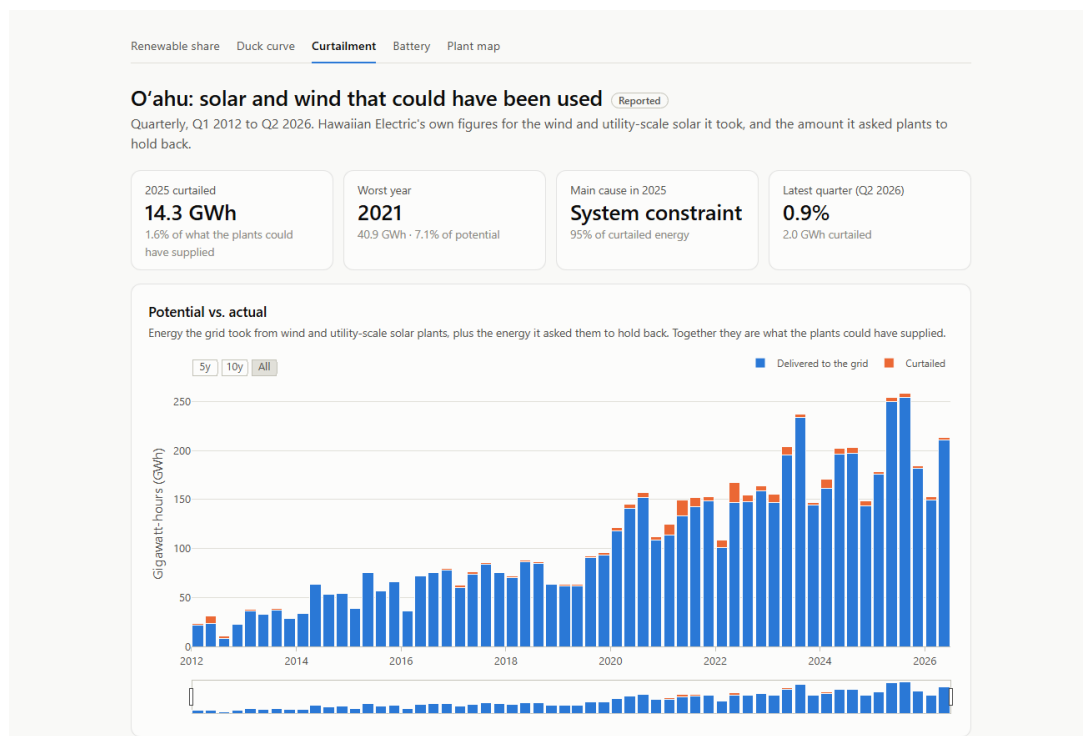


Figure 6: The Tier 3 page: 2025 curtailment, worst year, main cause, latest quarter, and quarterly delivered versus curtailed energy since 2012 (tagged “Reported”).

6.4 Battery simulation

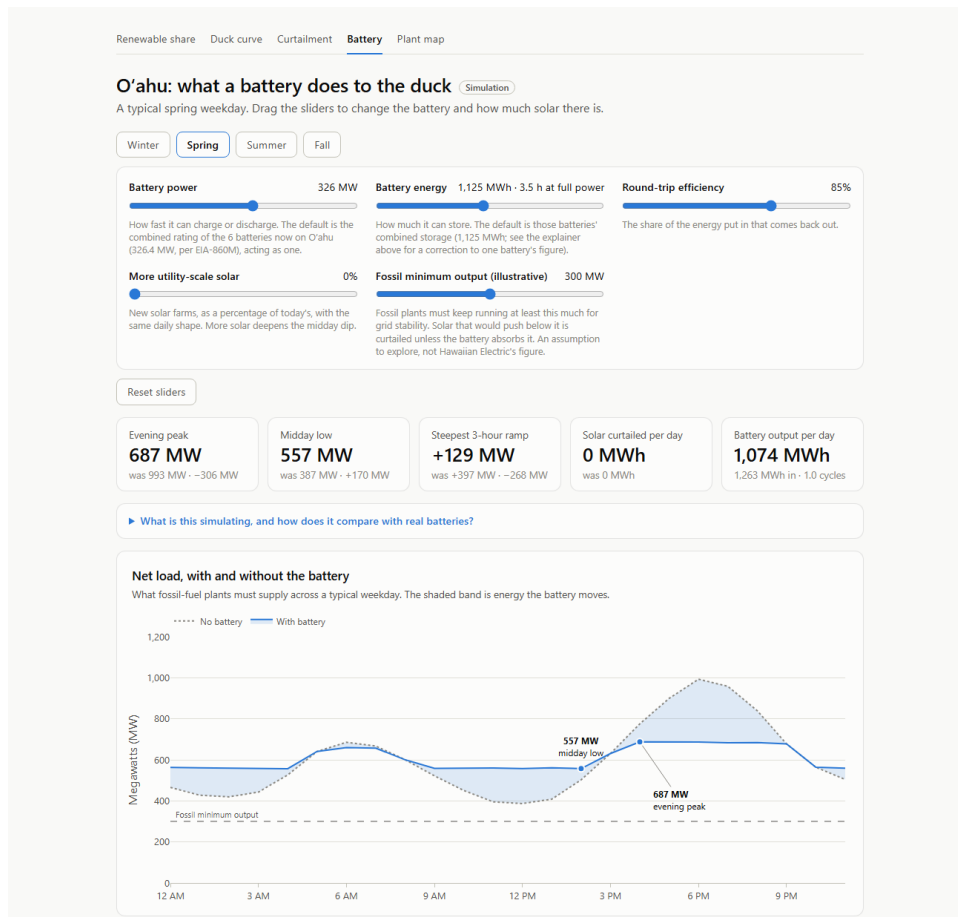


Figure 7: The Tier 4 page (spring, default sliders). With the island's whole battery fleet acting as one battery, the modeled evening peak falls from 993 to 687 MW and the midday low rises from 387 to 557 MW; the battery cycles about once a day.

The simulation is best read as a demonstration of the *interaction* between solar and storage rather than a forecast. With 200% more solar and no battery, about 783 MWh of solar a day would be curtailed against the 300 MW floor and the fleet absorbs all of it; the effect saturates further out (at +300% solar about 376 of 1,604 MWh/day are still curtailed; at +400% about 1,251 of 2,528). The real curtailed energy on the page (about 39 MWh a day on average in 2025) is shown only for scale.

6.5 The plant map

The map page (Figure 8) puts each unit on a time slider from December 1947 to April 2029; it opens on the current month, has a Today button, and Play sweeps the timeline month by month at about 109 months per second. Circle size is capacity in service on the selected date and colour is the fuel of the largest units; planned plants are faint circles with a dashed outline and are never counted as in service. The side panel and list update with the slider. In August 2022 the map shows 29 plants including the 203 MW AES coal plant; two months later the coal circle is gone, matching its retirement on 1 September.

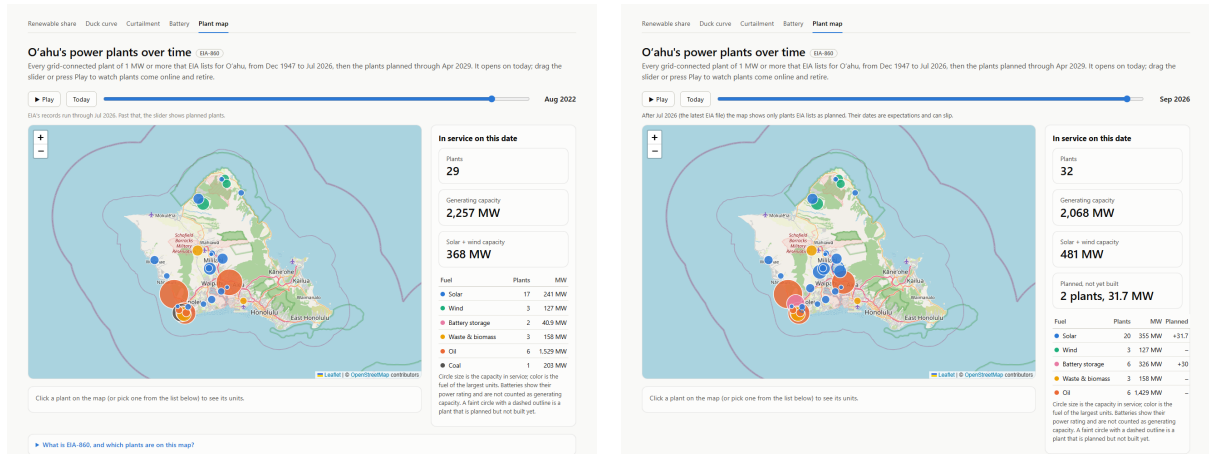


Figure 8: The plant map. Left: August 2022, the last month with coal (dark circle, 203 MW). Right: today's date (September 2026); the two dashed circles are solar-plus-battery projects that EIA lists as expected in September 2026, and the statistics count only in-service units.

7 Data quality, uncertainty and limitations

Table 7: Principal limitations, their effect and what mitigates or would fix them.

Limitation	Effect on what the charts show	Mitigation or possible fix
No public hourly grid data for Hawai'i	Tier 2 (and hence Tier 4) are modeled: shape from NREL simulations of 2018 weather, size from 2024 monthly totals; one average weekday per season	Labeled "Modeled"; peak checked against HECO's filing; real utility data would replace the shapes without changing the chart
State-to-island scaling	Rooftop solar is estimated for the state; O'ahu's share varies 0.65–0.83 by year and the home share (0.70) is population-based	Rooftop total anchored to HECO's reported quarterly totals; 0.77 only as fallback
1 MW reporting threshold and 2–3 month lag (EIA-923/860)	Small plants are invisible; newest months preliminary and read low	Shaded "preliminary" months; automatic correction on re-run
Wind and solar reported together (HECO)	Cannot say how much of curtailment was solar or wind	Ask for split by technology
Quarterly curtailment	Nothing about when in the day it happens or at which plant	Ask for hourly or event-level data
Category definitions unpublished	"System constraint" cannot be broken down, so what storage could relieve is unknown	Ask HECO for definitions; consult the utility's curtailment reports to the regulator
Plant list is configuration	A new or renamed plant must be added by hand (a discovery helper exists)	Automate by county match on each EIA-860M refresh
Definitions of "renewable"	Solar plus wind only; not the state RPS definition; waste-to-energy excluded	Stated on every page; a switch in the configuration
Retirements listed only from 2002	The map's early decades are sparse and survivor-biased	Explained on the page; would need a supplementary historical source
Modeled battery vs. real batteries	Load shifting only; perfect foresight; illustrative fossil minimum; no services	Labeled "Simulation"; explainer compares with real services
Planned-unit dates	Operators' expectations; cancelled projects vanish from the sheet; no planned retirements listed	Caveat shown with every planned plant

7.1 How errors were guarded against

- *Format drift.* Fetchers locate data by labels and fail with a clear message if a layout changes; fake-success pages are detected by content, not status code.
- *Silent unit and time-zone errors.* Tested (the 5-hour NREL shift; quarter/month mapping; strings that look like numbers; three apostrophe styles in “O‘ahu”).
- *Double counting.* Nested EIA rows are dropped; battery charging is excluded from totals.
- *Guarded corrections.* Overrides apply only to the exact value they were written for and are logged in the JSON.
- *Calculation changes.* Adding the rooftop line and later re-anchoring it were verified to leave every existing field unchanged, by comparing the outputs before and after.
- *Claims.* Two claims made earlier in the project were withdrawn after review: a statement that batteries “may not relieve” system-constraint curtailment (unsupported), and a report of a cause for the end of the early Kahuku wind farm that no source confirmed.

8 Additional data that would improve the visualizations

The current site is limited less by design than by the resolution of public data. Table 8 lists data that would most improve it, roughly in priority order, with the visualization it would enable and who would benefit. “Ask” means the data probably exists inside the utility or the regulator’s dockets and would have to be requested or found in public filings; we have not confirmed which are already published.

Table 8: Data that would most improve the visualizations, what it would add and who benefits.

Data	What it would add	Who would find it useful, and why
1. Hourly (or 5-minute) system load and generation by unit (utility system-operations data; “ask”)	Replace the modeled duck curve with measured net load; plot real days, extremes (highest peak, deepest belly) and trends by year; validate or correct the modeled overnight and midday levels	Utilities and planners (communication, integration studies); regulators (evidence for dockets); researchers (calibrating models); developers (sizing storage against real ramps); educators
2. Curtailment by hour, plant, technology and cause , with definitions of each reason (ask)	Show <i>when</i> curtailment occurs, which plants bear it, and separate solar from wind; a heat-map of curtailment by hour and month; test whether oversupply falls after batteries	Independent power producers and lenders (revenue risk under PPA terms, which vary by contract); storage developers (where a battery earns its keep); regulators (fairness of curtailment allocation)
3. Metered or inverter-telemetry rooftop solar and net demand , aggregated by hour and by circuit region (ask)	Replace the estimated rooftop layer; show real behind-the-meter output on clear and cloudy days; expose true customer demand; a reconciled energy balance	Utilities (visibility of rooftop solar is the hardest part of operating a high-DER grid); solar installers; policy analysts evaluating net-metering successor programs
4. Storage operations: hourly charge, discharge and state of charge per battery, and which services it provided	Compare the simulation with what batteries actually do; show service use (frequency response versus shifting); estimate real round-trip efficiency and cycling	Storage owners and investors; system planners; researchers testing dispatch models

Data	What it would add	Who would find it useful, and why
5. Customer-sited storage and flexible loads (home batteries, grid-service programs, EV charging, water heaters)	Add the evening supply that customer batteries provide and the demand-side flexibility that flattens the duck; EV charging load shapes	Program designers; regulators; households considering batteries; EV-infrastructure planners
6. Grid constraints: circuit or region hosting capacity, transmission limits, minimum-generation and reserve requirements, inertia	Explain <i>why</i> system-constraint curtailment occurs and how much room remains for new solar; overlay hosting capacity on the plant map; show the real minimum-output floor instead of the invented 300 MW	Developers (siting), utilities (interconnection planning), regulators, communities near proposed projects
7. Prices, costs and contracts: fuel prices, PPA or avoided-cost rates, tariffs (there is no wholesale market to draw on)	Convert MWh to dollars: the value of curtailed energy, the cost of the evening peak, a payback view of storage; time-of-use price overlays on the duck	Investors and developers (economics), ratepayer advocates, policy staff, storage vendors
8. Fuel consumption and emissions: EIA-923 also records fuel burned and heat input per plant (not yet used); emission factors	Heat rates, plant efficiency, estimated CO ₂ per MWh and its hourly marginal value; emissions avoided per MWh of solar; a carbon ledger	Sustainability officers, climate researchers, students, journalists; also plant owners benchmarking efficiency
9. Measured solar resource and weather: local irradiance, satellite-derived solar data, temperature	Replace “2018 weather shapes and a typical year” with actual days; explain variability (why one week’s share is low); capacity-factor analysis by year	Forecasters, developers doing yield assessment, researchers, utilities scheduling reserves
10. Reliability and events: outages, under-frequency load-shedding events, reserve shortfalls	Link the evening-peak stress to real events such as outages caused by generation shortfalls; a resilience view of when the system was tightest	Regulators, emergency planners, critical-facility operators, journalists
11. Other islands and the state (Maui, Hawai’i Island, Kaua’i) with local resources such as hydro and geothermal	Comparisons of renewable share, curtailment and storage by island; a statewide view; test of how well the architecture generalizes	State energy office, county governments, researchers; utilities benchmarking islands
12. Interconnection queue and project pipeline	A richer future view than EIA’s planned list: projects in the queue, their sizes, and probability of completion	Developers, planners, regulators, investors

8.1 Improvements possible with data already in hand

Some improvements need no new data, only new views of what is already cached:

- **Fuel-mix chart.** The Tier 1 JSON already carries generation for every fuel every month; a stacked area chart would show oil’s decline and waste-to-energy’s steady contribution alongside solar and wind.

- **Capacity-factor panel.** Combining EIA-860M capacity with EIA-923 generation (Table 6) gives per-plant and per-technology capacity factors, useful to owners and investors as a benchmark and to analysts as a data-quality check.
- **Fuel consumption and CO₂.** EIA-923 fuel data (collected on the same form) would allow a first-order emissions estimate and plant heat rates without new sources.
- **Retired-capacity and replacement view.** With unit dates, a chart of capacity added and retired per year (by fuel) would explain the steps in Tier 1.
- **Uncertainty display.** Shaded bands on the modeled duck curve (from the spread between individual weekdays in the NREL data) would show how much variation the average hides.
- **Scenario labels for storage.** A preset for “Kapolei alone” versus “fleet” would make the difference between the upper bound and the single-plant case explicit.

8.2 Why better data would matter to generation and supply companies

Different parties are exposed to different pieces of the picture:

- *Independent power producers and developers* are paid for energy delivered, so curtailment is direct revenue loss (the size of the exposure depends on each contract’s terms). Hourly, plant-level curtailment and cause data, together with the minimum-generation and hosting-capacity constraints, would let them value a site, choose between adding storage or more panels, and negotiate contract terms on evidence.
- *Storage developers* need the shape of the duck (ramps, midday surplus, evening peak) to size power versus duration, and evidence of which curtailment is storable. The gap between the simulation’s “upper bound” and reality (charging restrictions, services, degradation) is exactly what better data would close.
- *The utility* would gain little *information* but a lot of *communication*: a public, well-labeled version of its own system data makes the case for interconnection limits and investments understandable to customers and regulators.
- *Regulators and policy staff* need consistent definitions across programs (RPS compliance, net metering successors, curtailment allocation). A site that shows reported and modeled quantities side by side helps expose where a single number hides a definition choice.
- *Investors and lenders* price curtailment and capacity-factor risk; a benchmarked capacity-factor view and curtailment history by technology are direct inputs.

9 Novelty and usefulness: who this is for, and where it is and is not valuable

This assessment is candid and is based on our own knowledge of comparable public tools; we did *not* carry out a systematic survey, so a comparable Hawai‘i-specific tool that we are unaware of may exist.

9.1 What already exists

- **EIA’s Electricity Data Browser** and Open Data API publish the same plant-level generation (EIA-923) and inventory (EIA-860) data as tables and simple charts by state,

fuel or plant. It has no island-level view, no rooftop estimate anchored to a utility total, and no linking of sources.

- **EIA’s Hourly Electric Grid Monitor** [4] shows hourly load and generation by fuel, but for the Lower 48 balancing authorities; it does not cover Hawai‘i.
- **The California ISO’s daily outlook** shows a measured duck curve for California [9]. That is operationally superior data for its own system, but not for Hawai‘i.
- **Hawaiian Electric’s Key Performance Metrics pages** [5] publish the curtailment and renewable figures this project uses, as spreadsheets and PDFs without interactive views or cross-source comparisons.
- **Studies** by national laboratories and the utility on wind and solar integration in Hawai‘i exist as reports; they contain deeper analysis than any dashboard but are not living, explorable products.

9.2 Where the project is novel (modestly)

1. **Integration on consistent definitions for one island.** Generation history, an estimate of the part of supply EIA cannot see, curtailment by cause, the battery fleet and the plant timeline are on one site, with the definitions (solar plus wind only; 1 MW threshold; region as a plant list) stated where the reader needs them.
2. **A rooftop estimate anchored to a utility’s reported totals.** Combining a statewide monthly estimate with island-level quarterly measurements to reduce error from a fixed share of up to about 18% to rounding is a small but concrete technique, applied to every month, and documented.
3. **A modeled duck curve where none can be measured.** Calibrating simulated shapes to real monthly energy, checking against a regulatory filing, and labeling the result as modeled is a defensible way to illustrate the phenomenon for a grid that publishes no hourly data. It is not a discovery; it is a transparent stand-in.
4. **A battery what-if tied to the actual fleet.** The default is the island’s real fleet from EIA-860M, with a sourced correction to one filing, and results are put beside reported curtailment for scale.
5. **A time-travelling plant map with planned units.** Showing when each unit came online or retired, and what is expected, from a single official inventory, with the survivor-bias caveat spelled out.
6. **Provenance and reproducibility as features.** Every chart is tagged reported, modeled or simulation; the pipeline is tested, idempotent, re-runnable offline from a cache, uses no secrets in the repository, and is documented for learners. The tests caught real errors, and the project records the claims it withdrew.

9.3 Where it is not novel or not valuable

- *The data is not new.* Every number comes from a public source. A specialist with the same sources could reproduce any single chart in an afternoon.
- *The duck curve concept is over a decade old* and the modeled curve adds no new measurement. Operators and planners have measured data the site lacks; for them it is an illustration, not an analysis.

- *The battery optimizer is textbook* (dynamic programming on a quadratic cost) and stylized, so it offers no forecast and cannot support investment or dispatch decisions.
- *It is historical and diagnostic, not predictive.* No forecast of load, solar or prices; no real-time data; not a tool for scheduling, settlement or trading.
- *Coverage is narrow.* One island; no transmission or distribution detail; no prices; no emissions.
- *The value depends on trust in the labels.* A non-specialist could still misread a modeled chart as measured, which is why the “Modeled” tag and the plain-language caveats matter.

9.4 Target users and how well the site serves them

Table 9: Assessment of who benefits. “Usefulness today” is a judgement, not a survey result: the site has not been user-tested.

User	Usefulness today	Why, and what is missing
Students and educators (energy, engineering, data science, environmental studies)	High	Concrete, local, well-explained data; each concept (net load, curtailment, capacity versus energy) has an interactive example; the pipeline and tests are readable teaching material; missing: exercises and a data dictionary for the JSON (partly in the README)
Community members, journalists, policy and legislative staff	Medium–high	Clear answers to “how much of our electricity is renewable, and what is the catch?” and “why is solar being turned off?”; caveats stated; missing: emissions, cost and reliability context, and a plain-language summary page
Researchers (power systems, energy economics)	Medium	A tidy, provenance-tagged starting dataset and reproducible pipeline; missing: measured hourly data, so serious modeling still needs utility data
Developers, consultants and lenders (early screening)	Low–medium	Curtailment history, plant timeline and capacity-factor hints help screening; not sufficient for a project decision because curtailment is not by hour, plant or technology
Utilities and system operators	Low (as analysis), medium (as communication)	They hold better data; the site could serve as a public-facing explainer if fed their data
Regulators and energy offices	Low–medium	Useful for orientation and sanity checks of stated figures; not evidentiary

9.5 Recommended next steps

1. **Get the utility’s hourly data or curtailment detail** (Table 8, rows 1–2), by data request or from regulatory filings. This converts the two modeled and stylized pages into measured ones and is worth more than any other change.
2. **User-test with the intended audiences** (students, then journalists and staff) to check whether the “Modeled” and “Reported” labels are understood, since the central risk is misreading.
3. **Add the no-new-data views:** fuel mix, capacity factors, capacity added and retired by year, and an emissions estimate from fuel data.

4. **Generalize to other islands** using the region configuration, checking the assumptions that differ (hydro and geothermal on Hawai'i Island; a different utility on Kaua'i).
5. **Publish with versioned snapshots** so that a figure quoted in a report can be traced to a specific refresh.

10 Reproducibility and operation

Command	What it does
<code>python -m pipeline.run_pipeline</code>	Tier 1: fetch EIA-923 and small-scale solar, cache, process, write the renewable-share JSON
<code>python -m pipeline.run_curtailment</code>	Tier 3: fetch Hawaiian Electric's workbooks and write the curtailment JSON (also prints cross-checks; feeds the rooftop anchoring)
<code>python -m pipeline.run_duck_curve</code>	Tier 2: fetch NREL shapes and write the modeled duck-curve JSON
<code>python -m pipeline.run_plants</code>	Map and battery fleet: fetch the newest EIA-860M workbook and write the plants JSON
e.g. <code>python -m pipeline.run_plants --skip-fetch</code>	Rebuild from the cache with no network (the same flag works on the other run scripts)
<code>python -m pytest</code>	Python tests (109)
<code>node -test "tests/js/*.test.mjs"</code>	Battery optimizer tests (13)
<code>python tests/browser/run_browser_tests.py</code>	Headless-browser click-through tests of every page, including phone width
<code>python -m http.server</code> (inside web/)	View the site locally

Table 10: Main commands. The EIA and NREL APIs need free keys, stored in a git-ignored `.env` file.

The generated JSON files are committed so that the site works immediately after cloning; the raw SQLite cache is not. Everything in this report can be regenerated from the repository: the figures are browser screenshots of the site (saved in `report/figures/`) and the two data plots are drawn from the JSON values quoted in the tables. To build the report: `pdflatex report.tex` twice from the `report/` folder.

10.1 Data dictionary (main fields)

Table 11: The five JSON files the pages read.

File	Grain	Notable fields
<code>renewable_share.json</code>	One row per region and month	<code>period</code> , <code>generation_by_fuel_mwh</code> , <code>total_mwh</code> , <code>renewable_mwh</code> , <code>renewable_share_pct</code> , <code>provisional</code> , <code>rooftop_solar_mwh_est</code> , <code>rooftop_basis</code> (measured or assumed), <code>renewable_share_incl_rooftop_pct</code>
<code>duck_curve.json</code>	One row per region, season and hour	<code>customer_demand_mw</code> , <code>rooftop_solar_mw</code> , <code>grid_load_mw</code> , <code>utility_solar_mw</code> , <code>wind_mw</code> , <code>net_load_mw</code> ; region block with seasons' statistics and reference year
<code>curtailment.json</code>	Quarterly and annual rows per region	<code>delivered_mwh</code> , <code>curtailed_mwh</code> , <code>potential_mwh</code> , <code>curtailment_pct</code> , <code>reasons</code> (<code>oversupply</code> , <code>system_constraint</code> , <code>facility_requested</code>), <code>reasons_complete</code>

File	Grain	Notable fields
plants.json	One plant with its units; region summary	unit <code>online</code> , <code>retired</code> , <code>planned</code> , <code>status</code> , <code>mw</code> , <code>mwh</code> , <code>corrections</code> ; region <code>first_month</code> , <code>as_of</code> , <code>last_planned_month</code> , <code>battery_fleet</code>
milestones.json	Hand-written list of ten events per region	<code>month</code> , <code>date_text</code> , <code>title</code> , <code>text</code> , <code>source</code> , <code>url</code>

A Glossary

Ancillary services

Grid services besides bulk energy: frequency response, reserves, voltage support, black start.

Behind-the-meter (BTM)

Generation or storage on the customer’s side of the meter; the utility sees only net demand.

Black start The ability to restart a grid after a total blackout without outside power.

Capacity factor

Actual generation divided by the maximum possible at nameplate capacity over the same period.

Curtailement An instruction to a generator to produce less than it could.

DER Distributed energy resources: rooftop solar, customer batteries, flexible loads.

Duck curve The shape of net load on a sunny day: a midday dip and a steep evening climb.

EIA-860 / 860M

EIA’s annual (and monthly) inventory of generating units.

EIA-923 EIA’s survey of what each plant generated and the fuel it used.

Grid-forming inverter

An inverter that can set voltage and frequency (providing inertia-like behavior), unlike “grid-following” ones.

Inertia Resistance to frequency change from the rotating mass of synchronous generators.

MW, MWh, GWh

Megawatt (power); megawatt-hour and gigawatt-hour (energy).

Nameplate capacity

A unit’s rated maximum output.

Net load Demand minus wind and solar output: what dispatchable plants must supply.

Net metering

A policy crediting rooftop-solar owners for surplus power sent to the grid, historically at the retail rate.

Round-trip efficiency

The fraction of energy put into a battery that comes back out.

RPS

Renewable portfolio standard: a legal target for the share of renewable electricity.

Ramp rate How fast a generator or the system’s net load can change, in MW per hour.

State of charge

How full a battery is, in MWh or percent.

B Milestones marked on the renewable-share chart

#	When	Event	Date source
1	Mar 2011	Kahuku Wind starts (30 MW wind, 15 MW battery); first O’ahu wind in EIA’s data	EIA-860M
2	Feb 2012	Kapolei Solar Energy Park: first utility-scale solar in EIA’s data (Pearl City and Kalaeloa Solar Two follow in Nov–Dec)	EIA-860M
3	Nov 2012	Kawailoa Wind (69 MW), O’ahu’s largest wind farm	EIA-860M
4	Jan 2014	Honolulu Power Plant retires (104 MW oil steam, in service since 1954)	EIA-860M
5	Jun 2015	Act 97: 100% renewable electricity by 2045	[11]
6	Oct 2015	Net metering closes to new rooftop customers (cutoff 12 Oct)	[12]
7	Sep–Nov 2019	Four solar farms (Waipio 46, Kawailoa Solar 49, West Loch 20, Lanikuhana 15 MW; about 130 MW)	EIA-860M
8	Dec 2020	Nā Pua Makani wind (27.6 MW)	EIA-860M
9	1 Sep 2022	AES Hawai’i coal plant (203 MW) closes, the state’s last	EIA-860M; [13]
10	Dec 2023	Kapolei Energy Storage (185 MW / 565 MWh) starts	EIA-860M; [14]

Table 12: The milestones in `web/data/milestones.json`. The eight EIA-derived dates are checked against the plant data by a test.

References

- [1] U.S. Energy Information Administration. *Open Data API v2*. <https://www.eia.gov/opendata/>.
- [2] U.S. Energy Information Administration. *Form EIA-923, Power Plant Operations Report*. <https://www.eia.gov/electricity/data/eia923/>.
- [3] U.S. Energy Information Administration. *Form EIA-860 and EIA-860M, generator inventory*. <https://www.eia.gov/electricity/data/eia860/> and <https://www.eia.gov/electricity/data/eia860m/>.
- [4] U.S. Energy Information Administration. *Hourly Electric Grid Monitor*. <https://www.eia.gov/electricity/gridmonitor/>.
- [5] Hawaiian Electric. *Key Performance Metrics: Renewable Energy* (historical curtailment data). <https://www.hawaiianelectric.com/about-us/performance-scorecards-and-metrics/renewable-energy>.
- [6] Hawaiian Electric. *Adequacy of Supply*, report to the Hawai’i Public Utilities Commission, January 2021. https://puc.hawaii.gov/wp-content/uploads/2021/02/GEN-RPT.HECO_.ADEQUACY-OF-SUPPLY-2021.pdf.
- [7] National Renewable Energy Laboratory. *End-Use Load Profiles for the U.S. Building Stock* (Res-Stock and ComStock; public data on the Open Energy Data Initiative). <https://data.openei.org/>.
- [8] National Renewable Energy Laboratory. *PVWatts Calculator and API*. <https://pvwatts.nrel.gov/>.

- [9] California ISO. Material on the “duck curve” and daily outlook. <https://www.caiso.com/>.
- [10] Clearway Energy. *Waiawa* (operational project page). <https://www.clearwayhawaii.com/operational-projects-2/waiawa>.
- [11] Utility Dive. *Hawaii legislature sets 100% renewable portfolio standard by 2045*, 2015. <https://www.utilitydive.com/news/hawaii-legislature-sets-100-renewable-portfolio-standard-by-2045/394804/>.
- [12] pv magazine. *Hawaii shuts down net metering to new customers*, 14 October 2015. https://www.pv-magazine.com/2015/10/14/hawaii-shuts-down-net-metering-to-new-customers_100021550/.
- [13] Hawai‘i Public Radio. *Oahu dumps coal, rebounds with oil, longs for solar*, 1 September 2022. <https://www.hawaiipublicradio.org/local-news/2022-09-01/oahu-dumps-coal-rebounds-with-oil-longs-for-solar>.
- [14] Utility Dive. *Plus Power energy storage online in Hawaii (Kapolei Energy Storage)*. <https://www.utilitydive.com/news/plus-power-energy-storage-online-hawaii-HECO-rolling-blackouts/704561/>.
- [15] OpenStreetMap Foundation. *Tile usage policy*. <https://operations.osmfoundation.org/policies/tiles/>.